❖ STATES OF MATTER

→By considering all the recant and past studies till today, we are successful in finding around 22 states of matter.

→Humanity so much developed that we are able to create some states in our laboratories other than the natural ones.

→Here is the list containing all the 22 states of matter and below that is it's explanation or definition.

SR. No.	STATES OF MATTER	SR. No	STATES OF MATTER
1.	Solid	12.	Fermionic condensate
2.	Liquid	13.	Superconductivity
3.	Gas	14.	Superfluid
4.	Plasma	15.	Supersolid
5.	Excitonium	16.	Quantum spin liquid
6.	Electron degenerate	17.	String net liquid
7.	Neutron degenerate	18.	Dropleton
8.	Strange matter	19.	Time crystal
9.	Photonic matter	20.	Rydberg polarons
10.	Quantum spin hall	21.	Quark gluons plasma
11.	Bose-Einstein condensate	22.	Rydberg matter

Green -→naturally occurring states ; Blue→low temperature states
Red→high energy states

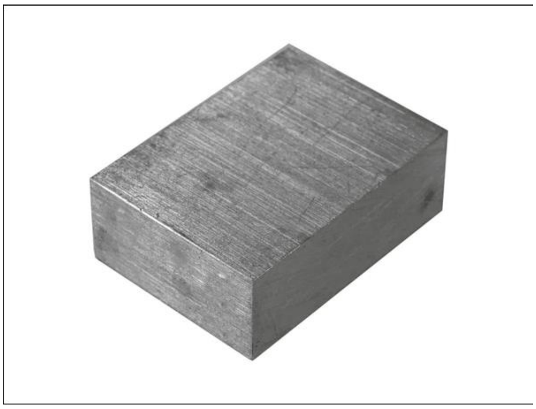
❖ Description of the above

1) **Solid**:- anything having definite shape and volume and their molecules are tightly bonded in known as solid.(Ex: Iron)
2) **Liquid**:- the substance who doesn't have any definite shape but do have definite volume(i.e. when pressure is increased volume doesn't change)is known as liquid.(Ex: water)
3) **Gas**:- material with indefinite shape and size, which takes the shape of the container it is put, is called gas.(Ex: air)
4) **Plasma**:- the ionized particles(like electron, proton, alpha particles),which can generate electric and magnetic field on its own, and the state in which the exist is called plasma.(Ex: fire)

5) **Excitonium**:- during a process known as Quantum Mechanical Paring, 'Excitons' are generated. This particles adds up to make the state called excitonium. This particles are integers which obey Bose statistics in the low-density limit.
6) **Electron degenerate**:- these state of matter is found in the core of the white dwarfs. In this state, electrons are strongly bonded to their respective atoms but they can be transferable to the adjacent atoms.
7) **Neutron degenerate**:- the neutrons found inside the neutron star go through the process of beta decay in which electrons are forced to bond with protons.
8) **Strange matter**:- this are also found inside the core of the neutron star. It is the one of the forms of quark matter with up, down and strange quarks which are assumed to occur in neutron star.
9) **Photonic matter**:- inside a quantum nonlinear medium, there are some photons which act like they have some mass and interact with one another and results in the formation of photonic molecule.
10) **Quantum spin hall**:- Theoretical phase of matter which is derived from the quantum hall state of matter.
11) **Bose-Einstein condensate**:- several bosons interact with each other and behave as a single wave/particle due to the same and collective quantum state of all bosons and this wave/particle state is known as bose-einstein condensate
12) **Fermionic condensate**:- when two fermions interact and start behaving like a boson, then this pair enters in the same quantum state without any kind of restriction and this state is called Fermionic condensate.
13) **Superconductivity**:- when certain substances when cooled to extremely low temperature(absolute zero temperature which is ≈-273.15˚c),there occurs a state of zero electrical resistivity and zero magnetic repulsion, then this state is known as superconductivity.
14) **Superfluid**:- when certain fluid is cooled at absolute zero temperature, they face zero resistance(viscosity≈0)against its flow, then this fluid is called superfluid.
15) **Supersolid**:- similar to superfluid, but here it will become more rigid in shape than its original rigidity all the time.

16) **Quantum spin liquid**:- a disorder state where quantum spins interact with each other and maintain a disorder to very low-temperature.
17) **String net liquid**:- this state of liquid have many closed loops which resembles as many closed string are tied in a net like shape. This obey the rule of branching.
18) **Dropleton**:- the quasiparticles which behave as a liquid is called Dropleton.
19) **Time crystal**:- a space-time crystal which changes its structure in space as well as in time
20) **Rydberg polarons**:- in this state of matter every atom is very large and are at low temperature.
21) **Quark gluons plasma**:- at very high energy state, quarks are also ionized and behave as plasma. This quarks are free to move in space independently.
22) **Rydberg matter**:- Rydberg atoms form excotic phase and this phase is known as Rydberg matter.

→**Here are images of above mentioned state of matter:-**

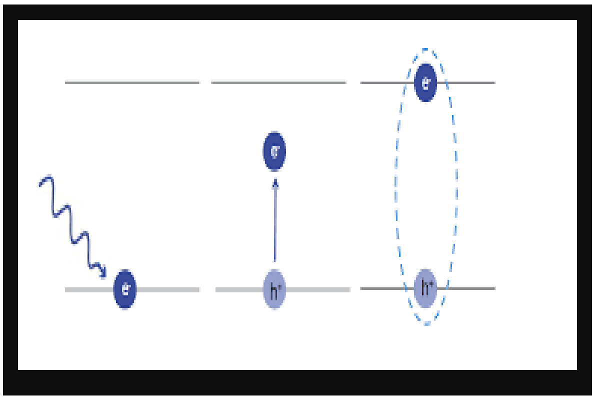


SOLID

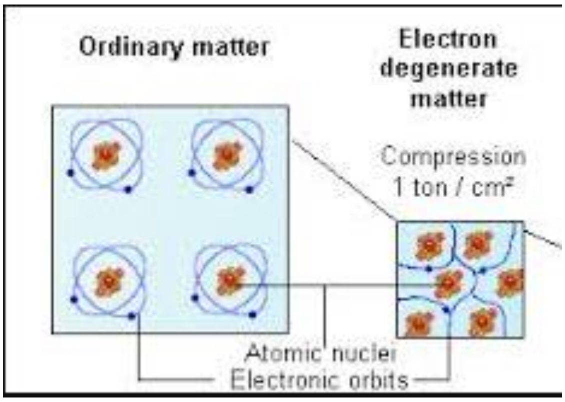
LIQUID

GAS

PLASMA

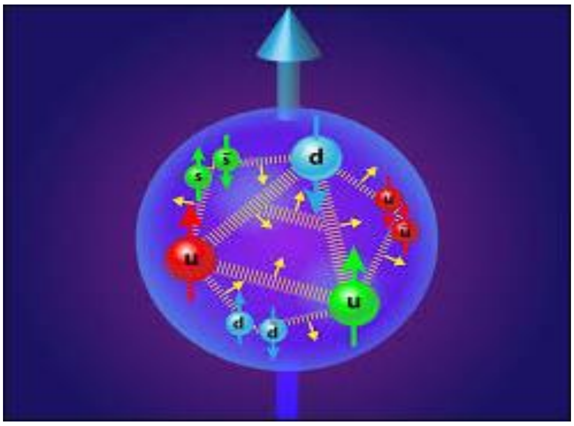


EXCITONIUM

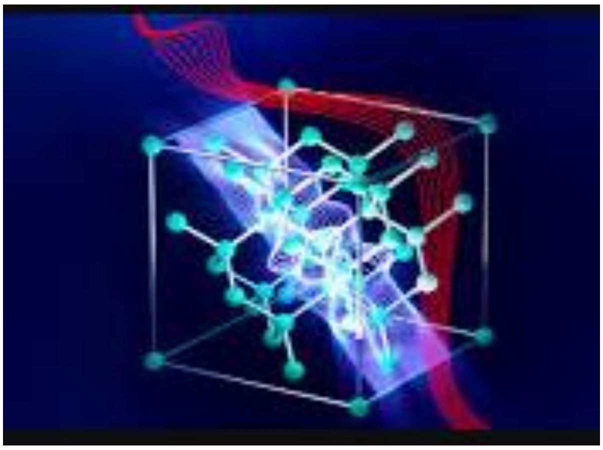


ELECTRON DEGENERATE

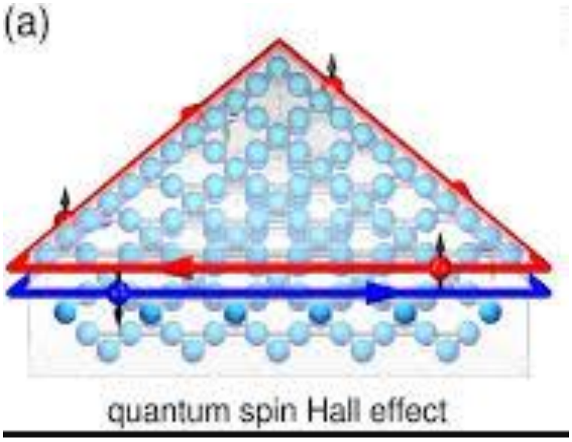
NEUTRON DEGENERATE



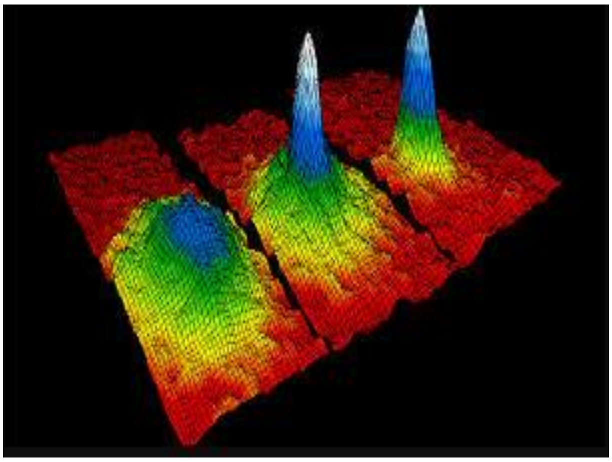
STRANGE MATTER



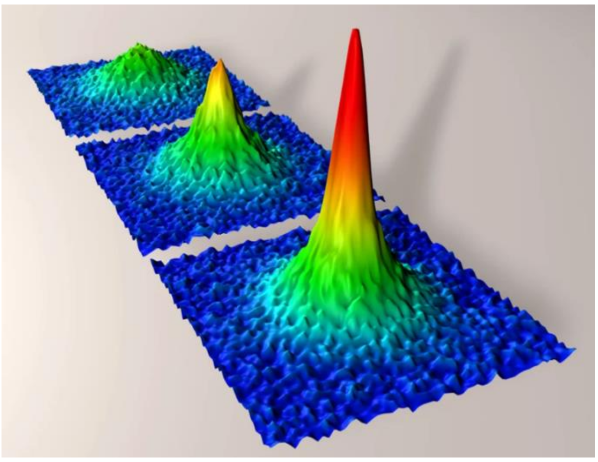
PHOTONIC MATTER



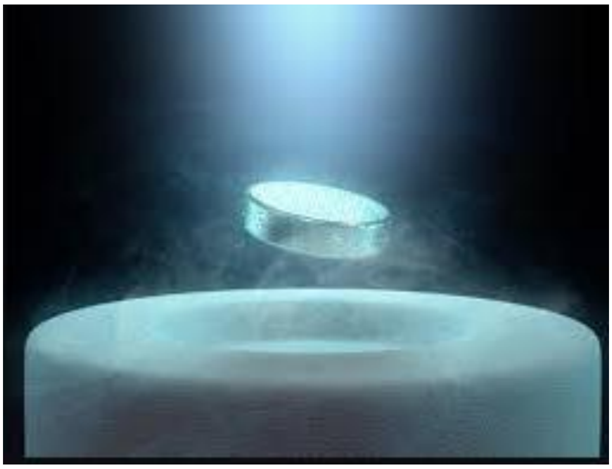
QUANTUM SPIN HALL



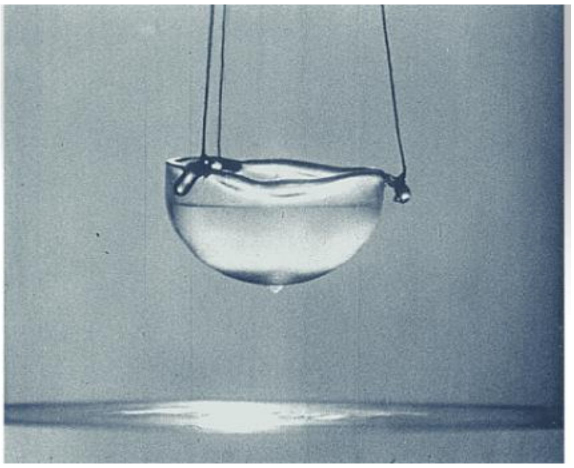
BOSE-EINSTEIN CONDENSATE



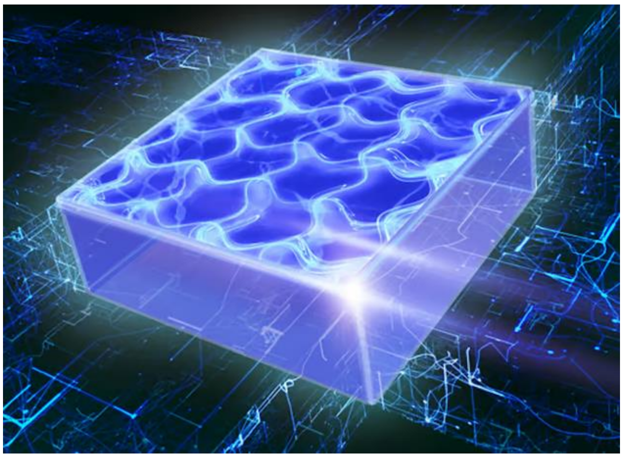
FERMIONIC CONDENSATE



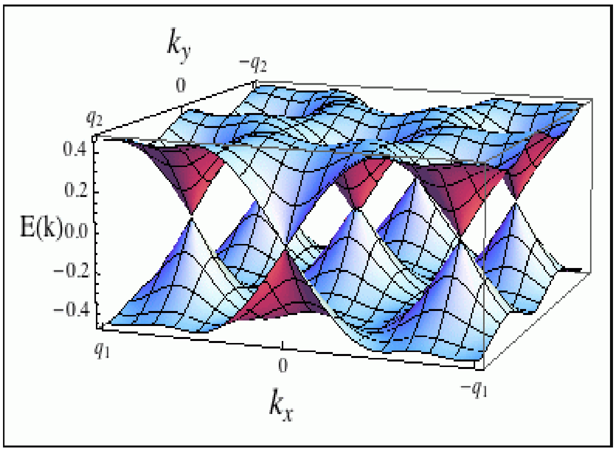
SUPERCONDUCTOR



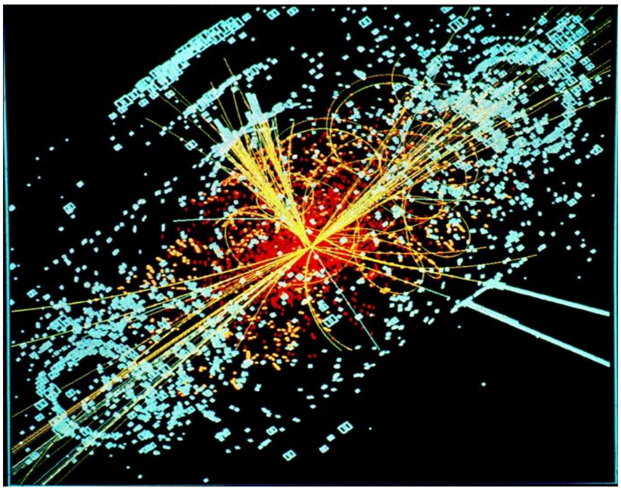
SUPERFLUID



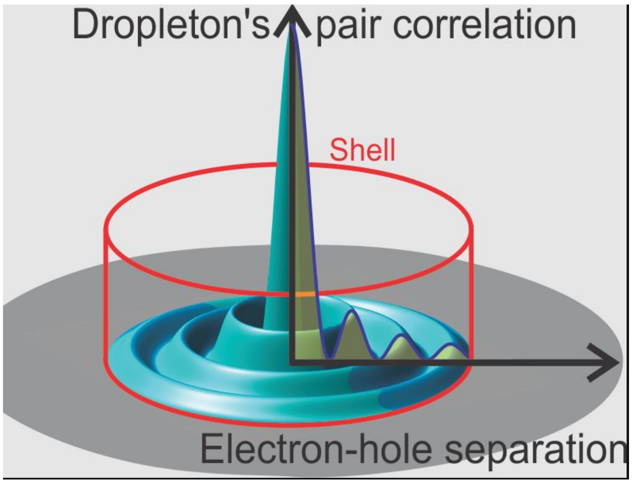
SUPERSOLID



QUANTUM SPIN LIQUID

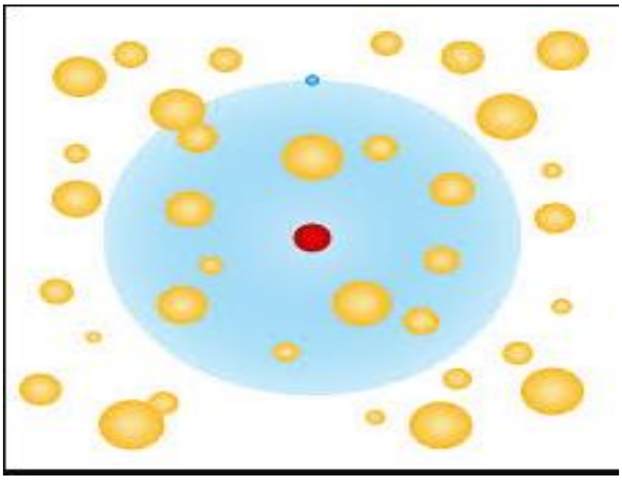


STRING NET LIQUID

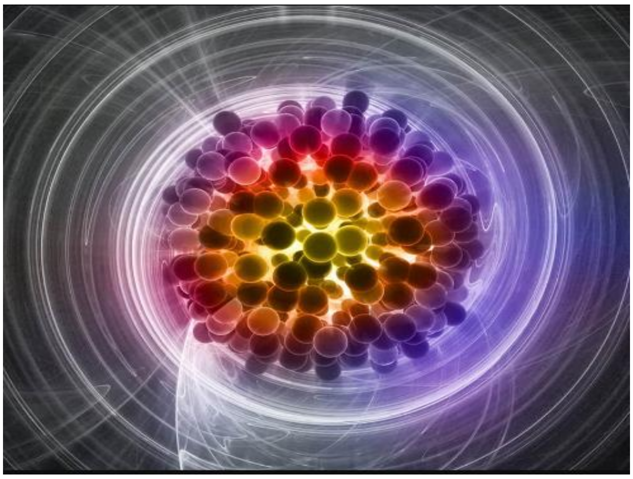


DROPLETON

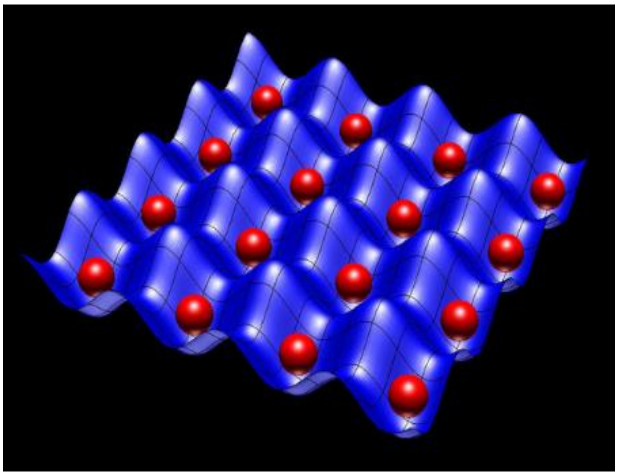
TIME CRYSTAL



RYDBERG POLARONS



QUARK GLUON PLASMA



RYDBERG MATTER

➔ ***<u>At last I would like share some fact on states of matter</u>.***

→Why understand states of matter?

- → Knowing a substance's state helps scientist to investigate its properties (toxicity, flammability, reactivity, etc.).
- → Understanding how matter behaves in different states allows scientist to formulate laws and principles that govern its behavior.
- → Different states of matter can have different levels of hazard.
- → Understanding these helps to establish safety guidelines.
- → Knowing the state of a substance is crucial for determining how to store and preserve it.

→Is there actually 22 states of matter?

- → The scientific community generally recognizes 5 distinct states of matter: solid, liquid, gas, plasma, bose-einstein condensate.
- → Other are intermediates or exotic states which occur during any process and are not distinctly recognized.

Thank you